

# Deep Learning: Final project

Retraining, Fine-Tuning and Grad-CAM  
Analysis of a VGG19-Based CNN

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## Can we trust the model?



### Interpretability

#### Grad-CAM

What does the model see?



### Robustness

#### Occlusion

What happens if key regions disappear?



### Generalization

#### RAF-CE

Does it transfer to new datasets?

# Datasets used

## CK+

Initial training



- Lab-controlled environment
- Posed facial expressions
- Used for baseline features

## FER 2013

Main training



- 35,887 grayscale images
- Unstructured "in the wild"
- 7 emotion categories

## RAF-DB

Generalization & fine tuning



- Highly diverse population
- Compound emotions
- Validates generalization

# Data Preprocessing Pipeline

## 1. OPTIMIZATION



**FER2013.csv**

600 MB | Grayscale



**HDF5 Dataset**

150 MB | Fast Access

## 2. IMAGE PREP

(28709, 2304)



(28709, 48, 48)

*Grayscale → 3-Ch RGB  
Replication for VGG19*

## 3. TRAINING PIPELINE

### Transformations

- RandomCrop
- HorizontalFlip
- Normalization

*Augmentation reduces  
overfitting*

## 4. TESTING

### TenCrop

Multi-crop TTA for  
stability



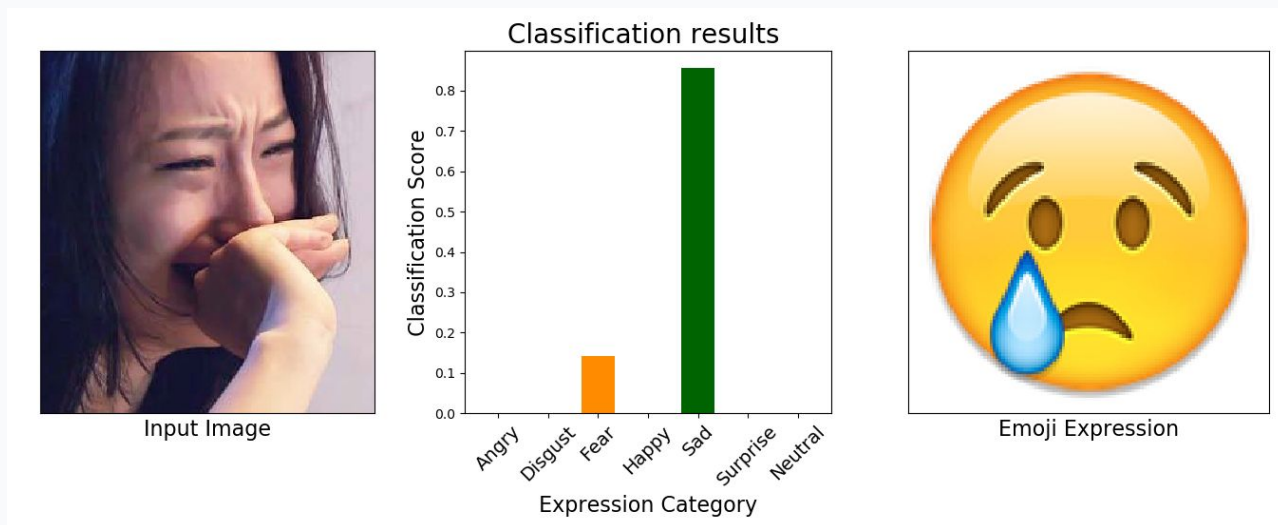
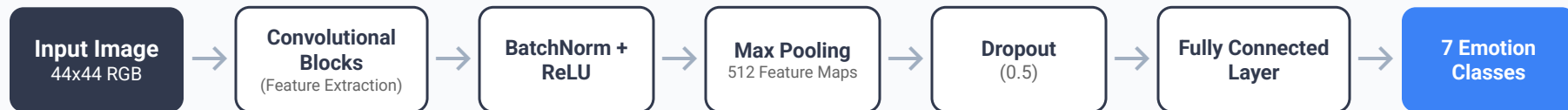
**VGG19**



Input: 48×48 RGB

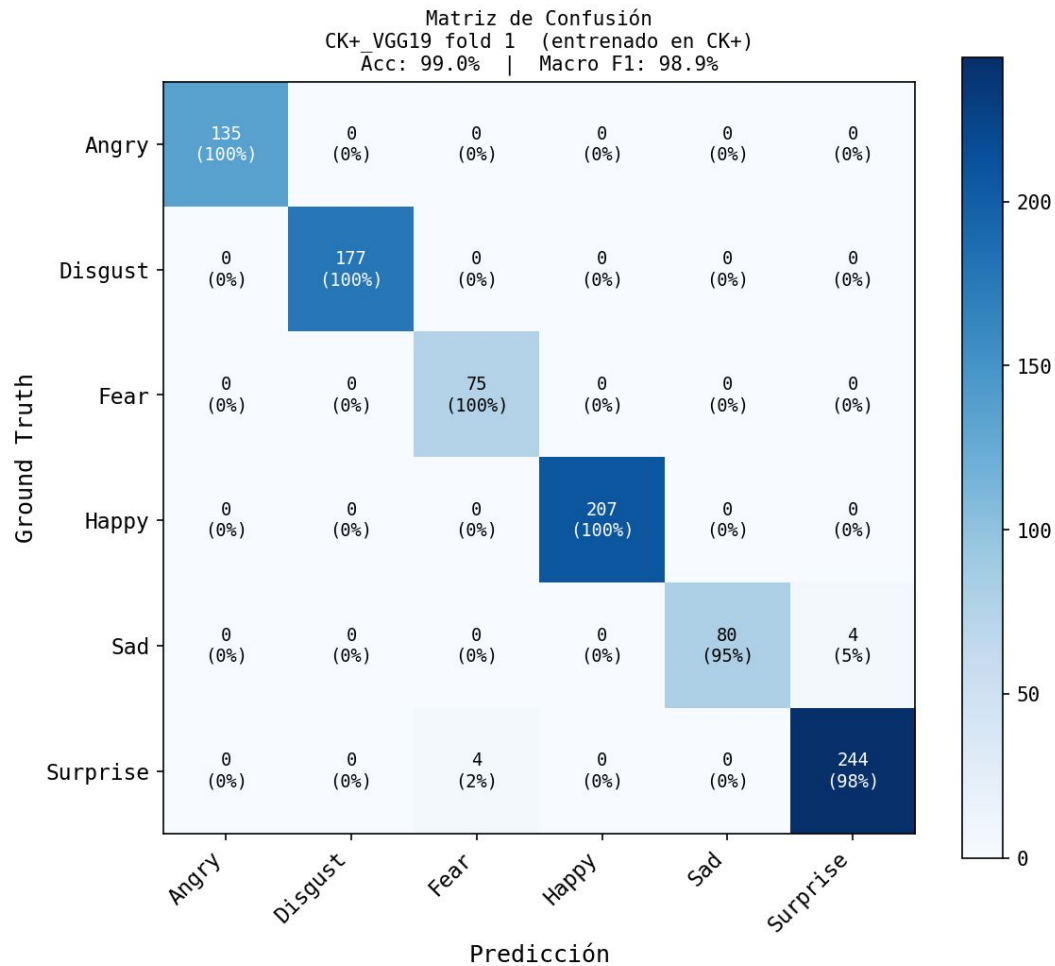
# VGG19 Architecture for Emotion Recognition

Based on the model by Wujie (2013)

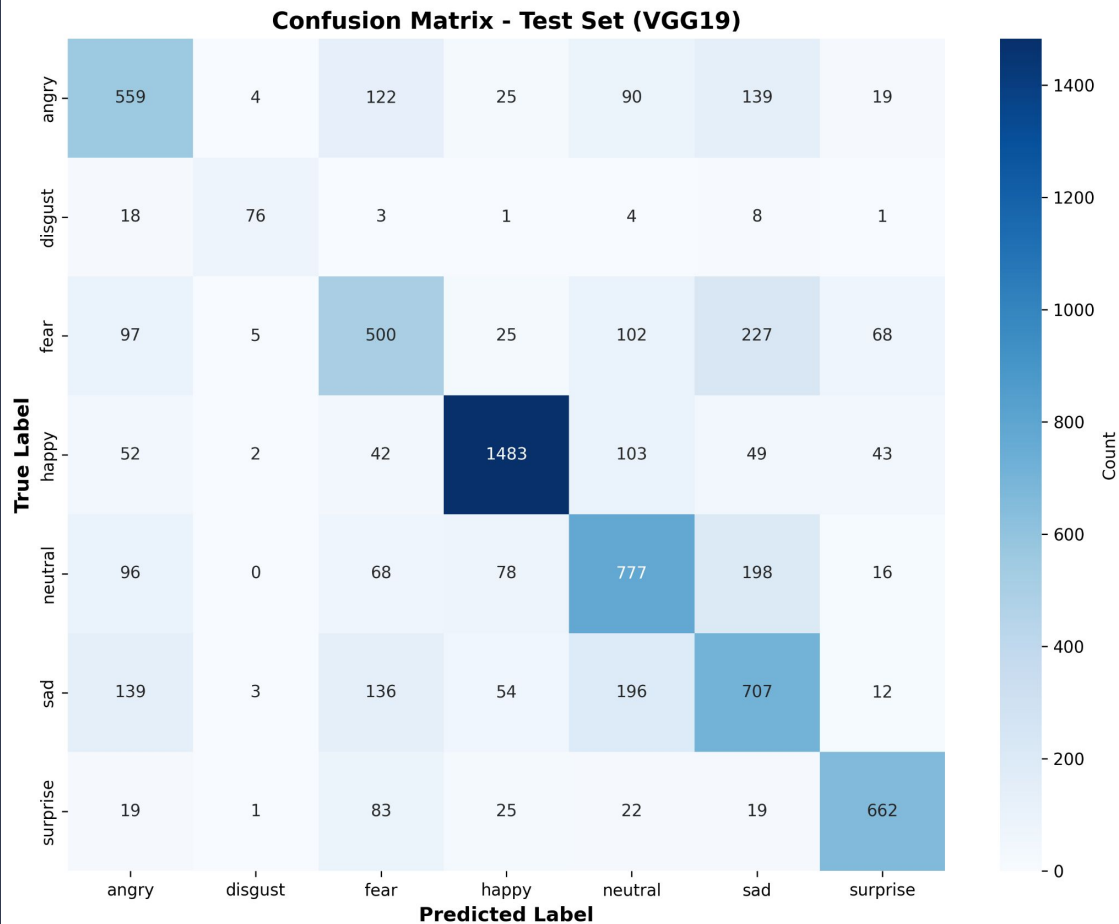


# Initial training

# Initial Training Phase: CK+



# Transition to FER

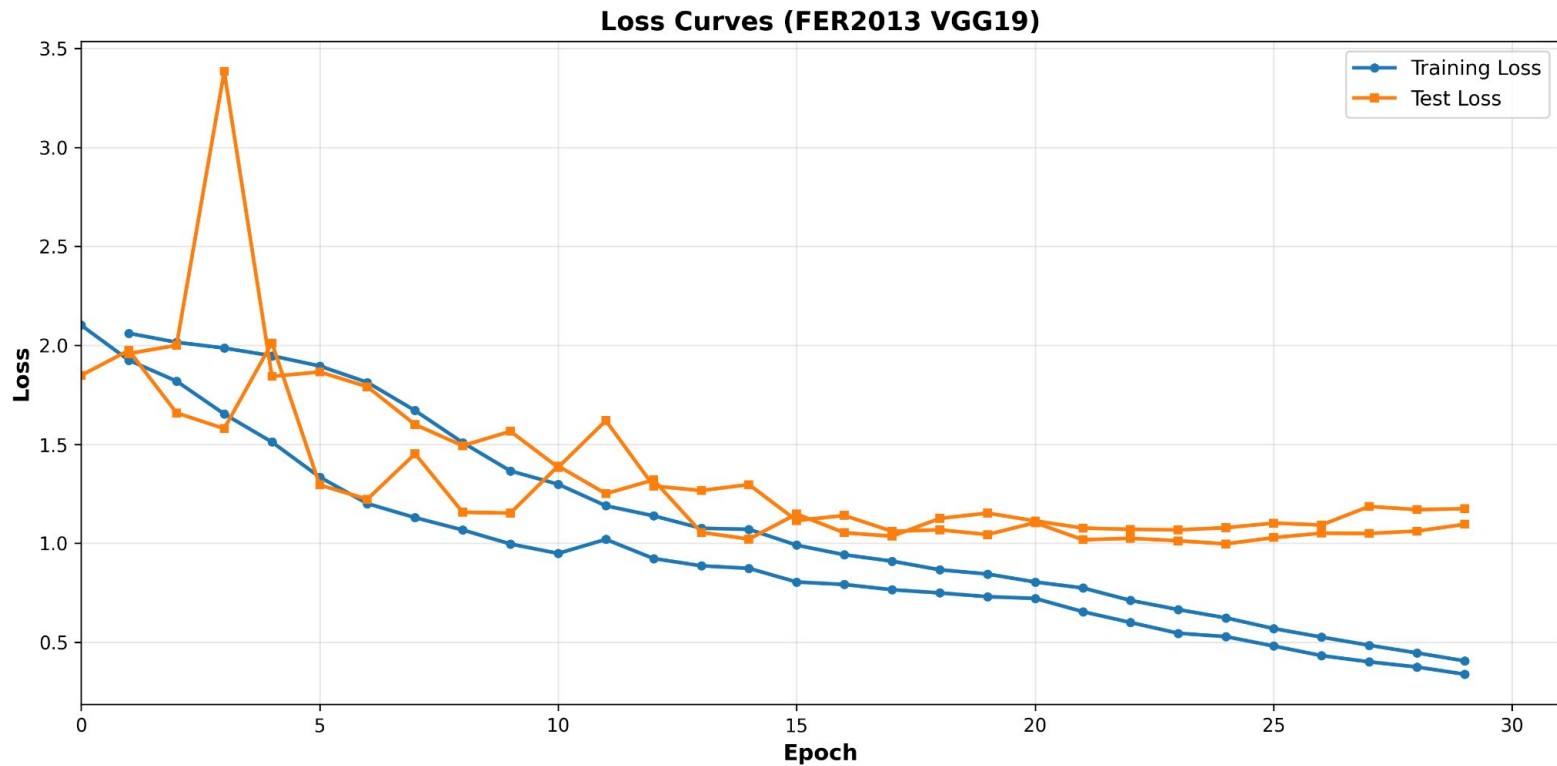




# Accuracy results

| Macro per - class accuracy: 72.47% |  | State of the art accuracy: 73.122% |  |
|------------------------------------|--|------------------------------------|--|
| Class                              |  | Accuracy                           |  |
| Angry                              |  | 65.58%                             |  |
| Disgust                            |  | 78.18%                             |  |
| Fear                               |  | 56.06%                             |  |
| Happy                              |  | 87.71%                             |  |
| Neutral                            |  | 59.43%                             |  |
| Sad                                |  | 85.58%                             |  |
| Surprise                           |  | 74.76%                             |  |

# Loss curve

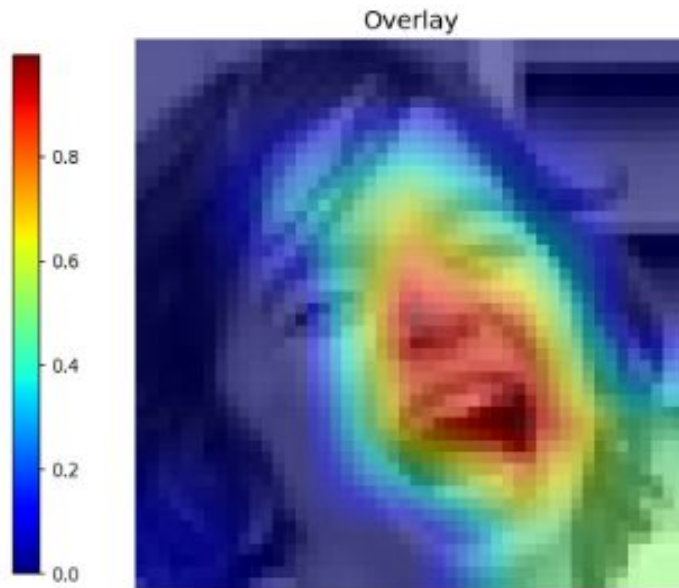


# Grad-CAM analysis

# What is Grad-CAM?

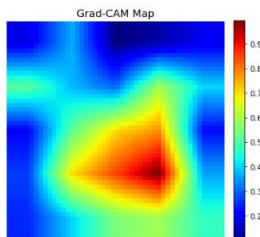
**Gradient-weighted Class Activation Mapping** is a technique used to produce "visual explanations" for decisions from CNN-based models.

- Identifies which pixels in an image are most important for a specific classification.
- Uses gradients of the target concept flowing into the last convolutional layer.
- Creates a localization map (heatmap) highlighting important regions.

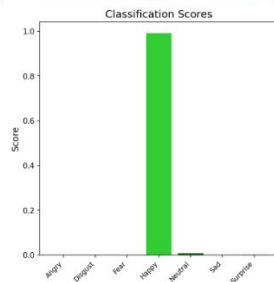


Example of Grad-CAM identifying key facial features

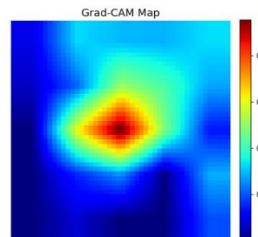
# GradCam analysis



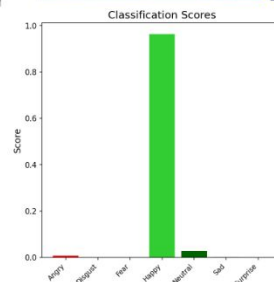
Ground Truth: Happy  
Predicted: Happy  
Match: ✓  
Confidence Scores:  
Angry: 0.0000  
Disgust: 0.0000  
Fear: 0.0001  
Happy: 0.9911  
Neutral: 0.0078  
Sad: 0.0001  
Surprise: 0.0000



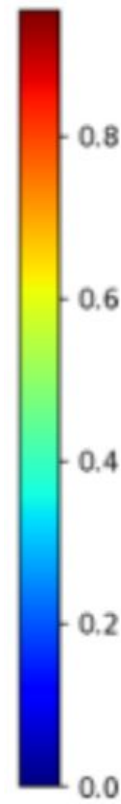
Happy: 0.99



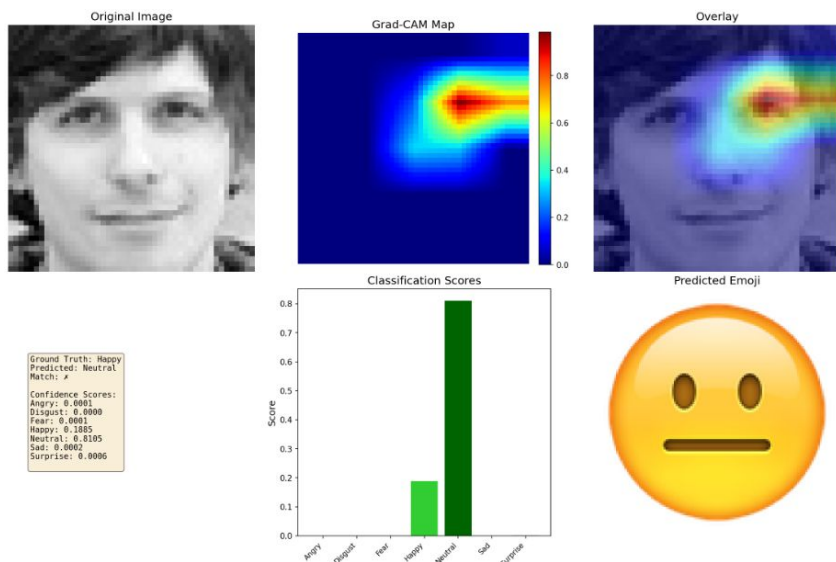
Ground Truth: Happy  
Predicted: Happy  
Match: ✓  
Confidence Scores:  
Angry: 0.0075  
Disgust: 0.0000  
Fear: 0.0001  
Happy: 0.9631  
Neutral: 0.0200  
Sad: 0.0013  
Surprise: 0.0001



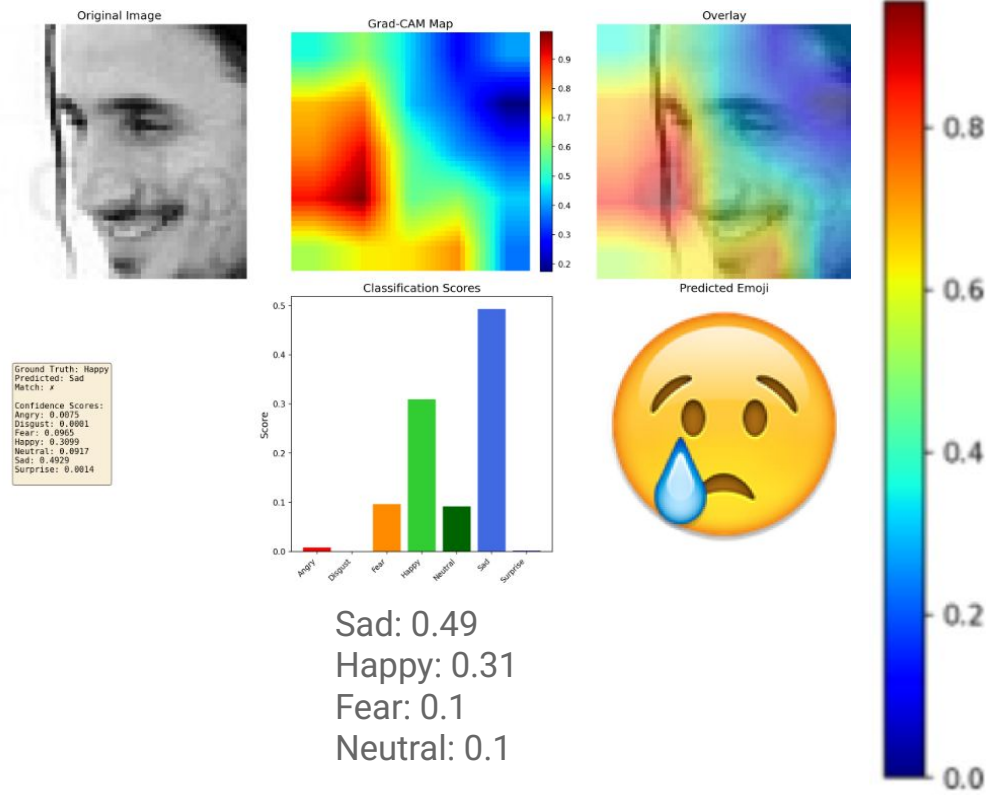
Happy: 0.96



# Incorrect cases



Neutral: 0.81  
Happy: 0.19

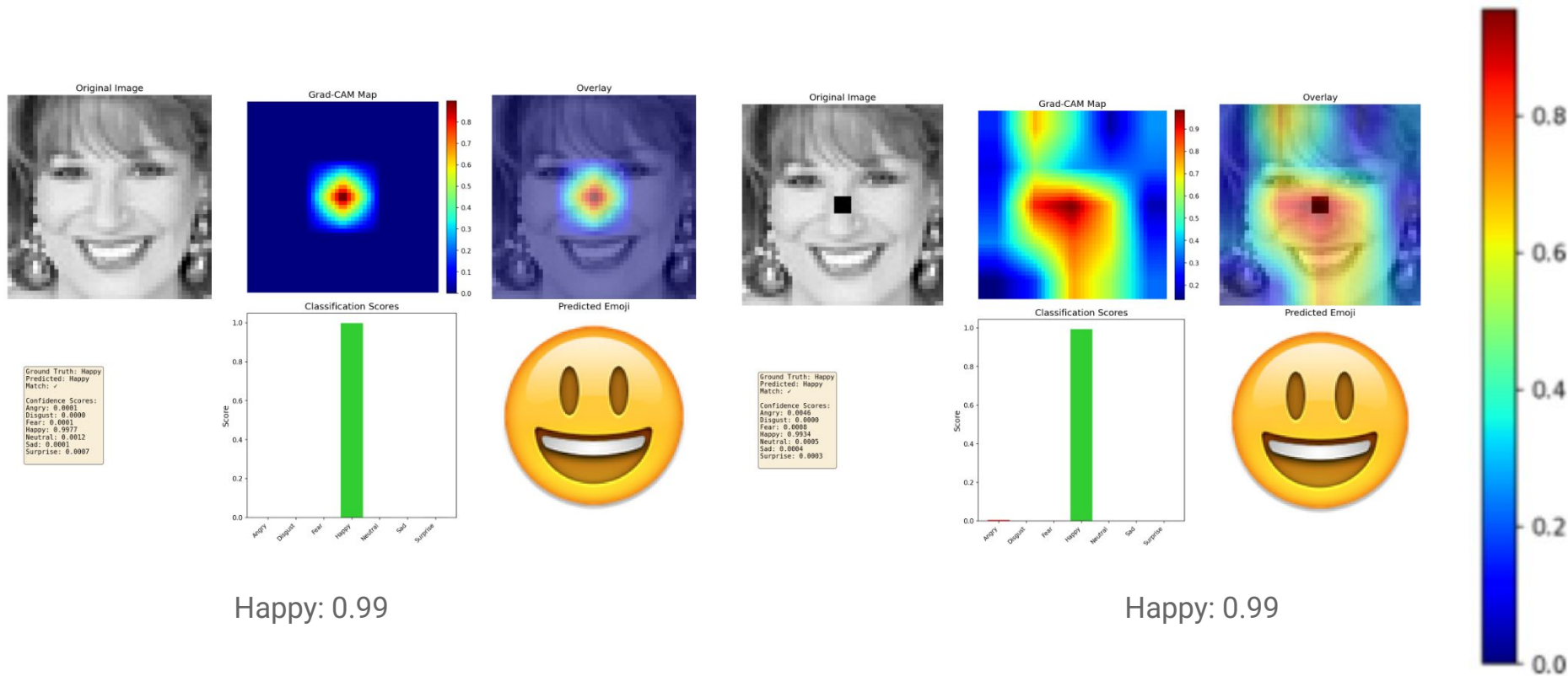


Sad: 0.49  
Happy: 0.31  
Fear: 0.1  
Neutral: 0.1

# Applying occlusion to our testing



# Interesting cases of the occlusion





# Fine - tuning with RAF-DB

# First testing with RAF-CE dataset

## Zero-Shot Evaluation

RAF-CE has 14 compound emotion classes. Labels mapped to closest FER emotion: "*Happily Surprised*" → "*Happy*".

## Results & Observations

- **Macro Accuracy:** 14.32% (random ~14%)
- **Uncertainty:** Loss 6.792 (highly uncertain)
- Model not trained for compound emotion blends
- **Confirmed need for fine-tuning**

## Results

| Class    | Accuracy |
|----------|----------|
| Angry    | 39.22%   |
| Disgust  | 7.36%    |
| Fear     | 17.95%   |
| Happy    | 35%      |
| Neutral  | 0%       |
| Sad      | 0.71%    |
| Surprise | 0%       |

# Fine – tuning

## Dataset Setup

- **Mixed:** FER2013 (28k) + RAF-CE (8k)
- **Sampling:** WeightedRandomSampler
- **Head:** 7-class label mapping

**WHY? Progressive adaptation**

**Avoid catastrophic forgetting**

## Training Pipeline

### PHASE 1

**Classifier Only:** lr=1e-3 (20 epochs)



### PHASE 2

**Last Conv Block:** lr=1e-4 (10 epochs)



### PHASE 3

**Full Fine-Tuning:** lr=1e-5 (10 epochs)

✓ **Best Checkpoint saved by combined validation accuracy**

**Objective:** Improve RAF-CE performance while **preserving FER2013 knowledge**

# Fine-tuning Performance Gain

## Before Fine-tuning

Macro Accuracy  
**14.32%**

Loss  
**6.792**

| Class    | Accuracy |
|----------|----------|
| Angry    | 39.22%   |
| Disgust  | 7.36%    |
| Fear     | 17.95%   |
| Happy    | 35%      |
| Neutral  | 0%       |
| Sad      | 0.71%    |
| Surprise | 0%       |



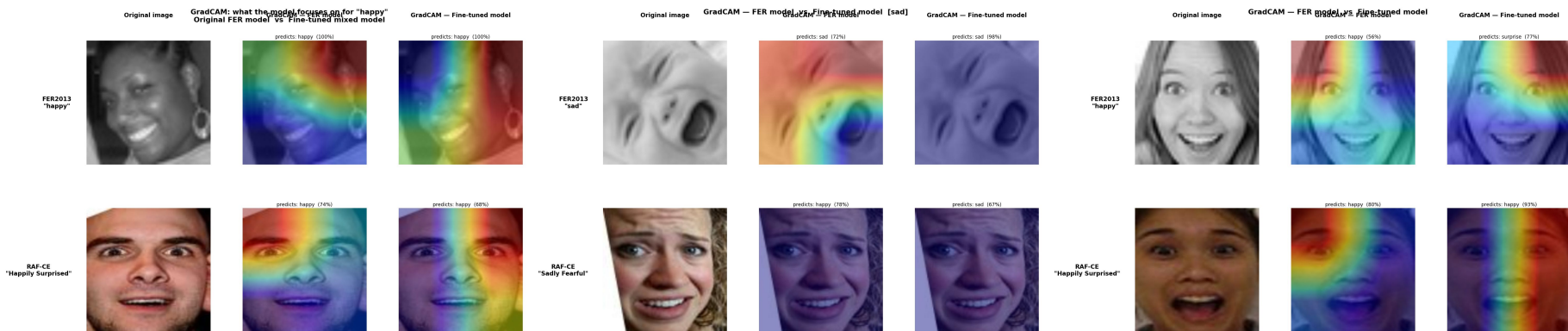
## After Fine-tuning

Macro Accuracy  
**30.31%**

Loss  
**1.732**

| Class    | Accuracy |
|----------|----------|
| Angry    | 15.69%   |
| Disgust  | 61.40%   |
| Fear     | 44.23%   |
| Happy    | 39.44%   |
| Neutral  | 0%       |
| Sad      | 50.88%   |
| Surprise | 0%       |

# Correct Cases

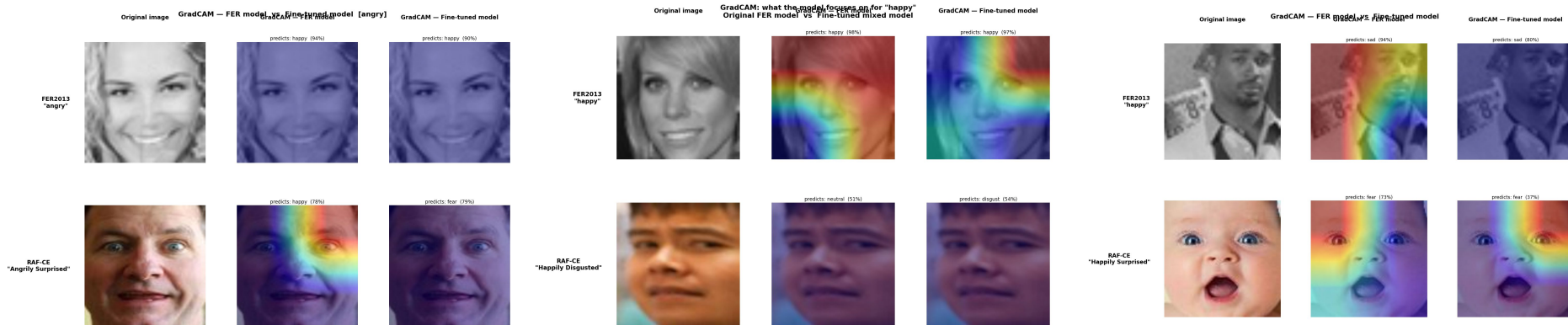


Changes in what the model observes.

The fine tuning works ideally.

Confidence improves.

# Incorrect Cases



The emotions are inherently ambiguous.

Mapped as Happy, but it's more disgust.

Incorrect mapping from the dataset.

# Discussions



## 1. Dataset quality strongly affects performance

- CK+ achieved high scores but generalized poorly
- FER2013 produced more realistic behavior



## 2. Emotion recognition is inherently ambiguous

- Several emotions share similar facial patterns
- Confusion matrix reflected human-like ambiguity



## 3. Grad-CAM improved interpretability

- Most predictions focused on meaningful facial regions
- Some unexpected attention patterns appeared



## 4. Robustness remains a challenge

- Occlusion experiments revealed sensitivity to perturbations
- CNNs can still rely on unstable visual cues

# Conclusions



## Model Adaptation

Adapting pretrained models proves effective for emotion recognition tasks.



## Careful Fine-tuning

Robust performance is contingent upon meticulous fine-tuning strategies.



## Data Integrity

Success requires balanced datasets and inherent robustness against environmental noise.



## Cross-Dataset Validation

Rigorous validation across diverse datasets is essential to ensure generalizability.



Thanks for your attention!  
Any questions?

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